B"H

NLP Deep Learning Project – Part 1

Authors: Bezalel Spolter and Yair Shmueli

Methodology

- Data Preparation and Tokenization
    - Making every lowercase and splitting sentences into individual words
    - Creating vocabulary of words and indexing based on the datasets received
- Embedding
    - GloVe – downloading and using pretrained embedding based on this method[1]
    - Word2Vec – training and utilizing this method on our dataset we received
- Model Definition – with 1 hidden layer, dropout layer and final linear layer which outputs 6 logits
    - LSTM (used on GloVe embedding)
    - GRU – bidirectional (used on Word2Vec embedding)
- Baseline Hyperparameter Configuration Run
    - Batch Size: 64
    - Learning Rate: 0.001
    - Hidden Dimension Size: 128
    - Dropout: 0.5 (high to help prevent overfitting on dataset)
    - Results: Classification report was used to make sure that precision and recall were considered in addition to accuracy, as distribution of classes of the datasets is not balanced between the classes. In addition, plots of accuracy and loss of train and validation as a function of epochs were output to check convergence time and if there was overfitting.

(These values were chosen because they were the middle of the group of values for each parameter which we wanted to check so they provide good baseline from which to deviate in tuning)

- Hyperparameter Tuning
    - Comparison of accuracy measure between the configurations and models. Accuracy was used for comparison because precision and recall are high enough that accuracy is not misleading for this unbalanced dataset.

[1] From here: https://nlp.stanford.edu/projects/glove/

B"H

Hyperparameter Tuning

For each method (embedding + DL model) after using a baseline for both models with the same hyperparameters, we ran 10 configurations where in each configuration a value was selected randomly for each hyperparameter from the options. The specific combinations of hyperparameter values which were chosen may be viewed in the notebook.

Note: comparison between values of hyperparameters' results must be taken with a grain of salt, as hyperparameters were not isolated when changed and combinations may be responsible for changes in results as opposed to single hyperparameter value change.

Hyperparameter	Options	Baseline	Best Result for GloVe + LSTM	Best Result for Word2Vec + GRU
Batch Size	32 , 64 , 128	64	64	128
Learning Rate	0.001 , 0.005 , 0.01	0.001	0.001	0.005
Hidden Dimension Size	64, 128, 256	128	256	256
Dropout	0, 0.3 , 0.5	0.5	0.3	0.3

Results (Baseline Result, Best of hyperparameter tuning):

Hyperparameter	Options	GloVe + LSTM	Word2Vec + GRU	Conclusion
Batch Size	32	0.906 0.89 0.886	0.926 0.925	There seem to be no great differences between batch sizes – for each model
	64	0.925 0.923 0.9075 0.899 0.893 0.89 0.876 0.873	0.921 0.92 0.924 0.912 0.912 0.905	
	128		0.929 0.926 0.928	
Learning Rate	0.001	0.925 0.923	0.921 0.92 0.925 0.912 0.912	For LSTM – lower learning rate seem to get

			0.905	better results, for GRU – higher learning rate seems to get better results
	0.005	0.906 0.9075 0.89 0.89 0.873	0.926 0.929 0.926	
	0.01	0.899 0.893 0.876	0.928 0.924	
Hidden Dimension Size	64	0.906 0.876 0.873	0.905	128 and 256 seem to be better results than 64, but differences between 128 and 256 are negligible
	128	0.925 0.9075 0.899 0.893	0.921 0.92 0.926 0.928 0.924 0.925	
	256	0.923 0.89 0.89	0.926 0.929 0.912 0.912	
Dropout	0	0.9075 0.893 0.89 0.876	0.925 0.912 0.905	Differences between dropout rates do not seem to be large – for the same model
	0.3	0.923 0.906 0.899 0.89 0.873	0.929 0.92 0.924	
	0.5	0.925	0.921 0.926 0.926 0.928 0.912	

Comparison Between the Models

Overall, Word2Vec and GRU did better (through different configurations) than GloVe and LSTM did in terms of accuracy. However, difference between best hyperparameter values (only batch size and learning rate are different in the best configuration of both models) and the best accuracy of this configuration are small between the models (0.925 for LSTM and 0.929 for GRU).